A Two-Way Higher-Order Helmholtz Propagator in Orthogonal Terrain-Following Coordinates: A Bremmer–Pseudodifferential Formulation

Mikhail S. Lytaev

Abstract

We formulate a two-way range-marching method for the scalar Helmholtz equation over irregular terrain. An orthogonal terrain-following map turns the boundary into a coordinate line and leaves an identity region above the coordinate transition. A Liouville transformation then writes the equation as $\varphi_{\xi\xi} + B(\xi)\varphi = 0$, where B is a symmetric vertical pseudodifferential operator. On a propagating spectral subspace, its positive square root defines flux-normalized forward and backward amplitudes. Their exact coupled evolution has diagonal one-way generators and a symmetric Bremmer interaction operator. Iterating the corresponding two-point integral equations gives successive reflection orders while preserving the distinction between local higher-order corrections and genuine backscatter. Weyl symbols identify the principal dispersion relation, the leading coupling, and the first commutator correction. We then describe a discrete realization that reuses the metric sampling and high-order vertical rational solves of an existing terrain-following propagator, while making clear which matrix fractional-power and two-boundary components a two-way implementation still requires. This paper gives the formulation and an implementation blueprint; it does not report numerical results for a completed two-way solver.

1 Introduction

One-way parabolic-equation (PE) solvers are effective when the field is predominantly directed along a preferred range coordinate. They omit the backward-going component, however, and cannot represent reflections generated by range variation except through errors or ad hoc corrections. Bremmer's coupling series provides a systematic alternative: decompose the field into forward and backward waves, propagate each component with a one-way operator, and couple the two components through an interaction operator [1, 2, 3].

For irregular terrain, the coordinate choice matters to every part of this construction. Staircase boundaries introduce artificial range discontinuities; shift maps retain a mixed derivative in the full Helmholtz equation. We use an orthogonal terrain-following map of the type developed for the high-order PE in [8]. It straightens the lower boundary, has no mixed derivative, and becomes the identity above a finite transition layer. This geometry gives a particularly direct setting for a two-way operator factorization.

The result below is an analytical and numerical-method formulation, not a claim that the current JAX terrain-following code already solves the two-way problem. That code advances one range direction. A two-way solver additionally needs discrete operator roots, a backward sweep, and a two-point range boundary solve. Our aim is to specify those pieces in a form that reuses the existing terrain metric and high-order vertical discretization.

Scope. We treat a time-harmonic scalar field at fixed frequency, with constant density in the derivation and a real, smooth wavenumber $k(\xi, \eta)$. The propagating part of the spectrum is assumed

to stay away from a turning point. Loss, variable density, interfaces, and evanescent modes can be included in the discrete model, but they remove some of the self-adjointness and flux-unitarity statements made here. Those extensions are identified where relevant.

2 Orthogonal terrain-following Helmholtz equation

Let (ξ, η) denote orthogonal coordinates with scale factors $h_\xi(\xi, \eta) > 0$ and $h_\eta(\xi, \eta) > 0$. Write $r = h_\eta / h_\xi$. In two dimensions,

$$\Delta\psi = \frac{1}{h_\xi h_\eta} \left[\partial_\xi (r \partial_\xi \psi) + \partial_\eta (r^{-1} \partial_\eta \psi) \right]. \tag{1}$$

The map is chosen so that $\eta = 0$ follows the terrain (or seabed), ξ labels orthogonal trajectories, and $h_\xi = h_\eta = 1$ in an exterior identity region. The construction in [8] obtains the level curves from a smoothed harmonic extension of the boundary and follows them with orthogonal trajectories. In particular, it avoids a mixed ξ–η derivative in (1).

The scalar Helmholtz equation $\Delta\psi + k^2\psi = 0$ is therefore

$$\partial_\xi (r \partial_\xi \psi) + \partial_\eta (r^{-1} \partial_\eta \psi) + h_\xi h_\eta k^2 \psi = 0. \tag{2}$$

Apply the Liouville scaling

$$\psi = r^{-1/2} \varphi. \tag{3}$$

After division by $r^{1/2}$, equation (2) becomes

$$\partial_\xi^2 \varphi + B(\xi)\varphi = 0, \tag{4}$$

where the operator in η is

$$B\varphi = r^{-1/2} \partial_\eta \Big[r^{-1} \partial_\eta (r^{-1/2} \varphi) \Big] + \big(k^2 h_\xi^2 - V \big) \varphi, \qquad V = \frac{\partial_\xi^2 (r^{1/2})}{r^{1/2}}. \tag{5}$$

For real coefficients and compatible self-adjoint vertical boundary conditions, $B(\xi)$ is symmetric in $L^2(d\eta)$. The term V is an explicit range-direction potential. The vertical divergence form in (5) must be retained in a discretization to preserve symmetry and the associated flux identity.

The orthogonal map has an identity exterior. This permits the same physical outgoing or incoming vertical boundary condition to be imposed above the terrain transition as in a flat computational domain. It also localizes the range variation of the metric, a useful property for both the Bremmer coupling and its numerical truncation.

3 Weyl symbol and local dispersion

We use $D_\eta = -i\partial_\eta$ and let q be its dual variable. For a symbol $a(\xi, \eta, q)$, Weyl quantization is

$$\mathrm{Op}^w(a) u(\eta) = \frac{1}{2\pi} \iint e^{i(\eta - \eta')q} a\left(\xi, \frac{\eta + \eta'}{2}, q \right) u(\eta')\, d\eta'\, dq. \tag{6}$$

The Weyl product has expansion

$$a \# c = ac + \frac{i}{2}\{a, c\} - \frac{1}{8} \left(a_{qq} c_{\eta\eta} - 2 a_{q\eta} c_{q\eta} + a_{\eta\eta} c_{qq} \right) + \cdots, \qquad \{a, c\} = a_\eta c_q - a_q c_\eta. \tag{7}$$

Thus the symbol of a commutator starts with $i\{a, c\}$, while the first derivative term cancels for the Weyl symbol of a square.

For $D = -i\partial_\eta$ and smooth g, the identity

$$DgD = \mathrm{Op}^w\left(gq^2 + \frac{1}{4}g_{\eta\eta}\right) \tag{8}$$

gives the exact Weyl symbol of (5):

$$B = \mathrm{Op}^w(b), \qquad b = k^2 h_\xi^2 - \frac{q^2}{r^2} - V - \frac{r_\eta^2}{4r^4}. \tag{9}$$

The principal symbol is

$$b_0 = k^2 h_\xi^2 - \frac{q^2}{r^2}. \tag{10}$$

The characteristic equation of (4) is $\kappa^2 = b_0$, with κ dual to ξ. In terms of the physical vertical wavenumber q/h_η and range wavenumber κ/h_ξ, this is

$$\left(\frac{\kappa}{h_\xi}\right)^2 + \left(\frac{q}{h_\eta}\right)^2 = k^2. \tag{11}$$

For a ray angle θ measured from the level curve, $q = kh_\eta \sin\theta$ and $\sqrt{b_0} = kh_\xi \cos\theta$. The forward and backward characteristic sheets are separated by $2\sqrt{b_0}$ wherever the propagating aperture stays away from grazing incidence.

If metric coefficients vary on length ℓ and the characteristic wavenumber is k, the semiclassical parameter is $\varepsilon = (k\ell)^{-1}$. The two zeroth-order corrections in (9), V and $r_\eta^2/(4r^4)$, are of relative order ε^2 for a smoothly varying metric. The exact symbol is real; it has no order-ε imaginary subprincipal part in this Liouville gauge.

4 Exact forward–backward splitting

On a spectral subspace where $B(\xi)$ is positive and bounded away from zero, define its positive roots

$$\Lambda = B^{1/2}, \qquad Q = B^{1/4}, \qquad Q^2 = \Lambda. \tag{12}$$

All powers here are operator powers at the same ξ. Decompose the field and its range derivative as

$$\varphi = Q^{-1}(u_+ + u_-), \qquad \varphi_\xi = iQ(u_+ - u_-), \tag{13}$$

or, equivalently,

$$u_\pm = \frac{1}{2}\left(Q\varphi \mp iQ^{-1}\varphi_\xi\right). \tag{14}$$

The signs are fixed so that for a constant B, u_+ evolves as $e^{+i\Lambda\xi}$ and u_- as $e^{-i\Lambda\xi}$.

Differentiate (14), use (4), and retain the ξ-dependence of Q. This gives the exact coupled system

$$\partial_\xi \begin{pmatrix} u_+ \\ u_- \end{pmatrix} = \begin{pmatrix} i\Lambda + T & \mathcal{R} \\ \mathcal{R} & -i\Lambda + T \end{pmatrix} \begin{pmatrix} u_+ \\ u_- \end{pmatrix}, \tag{15}$$

with

$$T = \frac{1}{2}[Q_\xi, Q^{-1}], \qquad \mathcal{R} = \frac{1}{2}(Q_\xi Q^{-1} + Q^{-1}Q_\xi). \tag{16}$$

For real self-adjoint Q, $T^* = -T$ and $\mathcal{R}^* = \mathcal{R}$. Hence

$$\partial_\xi \left(\|u_+\|^2 - \|u_-\|^2 \right) = 0. \tag{17}$$

The invariant is the net range-directed flux. The individual squared norms are the forward and backward fluxes in this normalization.

Equation (15) is exact on the chosen positive spectral subspace. The one-way approximation sets $u_- = 0$ and discards the feedback through $\mathcal{R}$. The two-way model retains it. If B has evanescent or nonpositive eigenvalues, the positive-root construction and the unitary flux interpretation do not apply to those modes. A practical implementation must either separate them from the propagating aperture, use a complex branch with appropriate incoming/outgoing boundary conditions, or work with an alternative first-order state-space factorization.

5 Pseudodifferential expansion of the coupling

Let $\lambda = b^{1/2}$ on the propagating symbol region. Functional calculus gives, to principal Weyl order,

$$\sigma(\Lambda) = \lambda + O(\varepsilon^2), \qquad \sigma(Q) = b^{1/4} + O(\varepsilon^2). \tag{18}$$

The first correction to the square-root symbol follows directly from $\lambda_W \# \lambda_W = b$. With $\lambda_0 = \sqrt{b}$,

$$\sigma(\Lambda) = \lambda_0 + \lambda_2 + O(\varepsilon^4), \qquad \lambda_2 = \frac{(\lambda_0)_{\eta\eta}(\lambda_0)_{qq} - (\lambda_0)_{\eta q}^2}{8\lambda_0}. \tag{19}$$

This term records the phase-space curvature of the dispersion surface. It can be included in the diagonal phase when the leading-symbol rational approximation is not accurate enough; equivalently, one can construct a corrected symbol of L_β from the same operator functional calculus. The anticommutator in (16) has the principal symbol

$$\rho := \sigma(\mathcal{R}) = \frac{b_\xi}{4b} + O(\varepsilon^3), \tag{20}$$

where the displayed term is order ε relative to λ. The first Poisson-bracket contribution cancels in this anticommutator. For the commutator,

$$\sigma(T) = \frac{i}{32} \frac{\{b, b_\xi\}}{b^2} + O(\varepsilon^3), \tag{21}$$

which begins at order ε^2 relative to the principal one-way rate. The commutator is skew-adjoint and contributes a phase/transport correction to each directional component.

At leading order the interaction is therefore the logarithmic range variation of the local longitudinal wavenumber:

$$\rho \simeq \frac{1}{4} \partial_\xi \log b_0. \tag{22}$$

For a narrow ray packet this reduces to the corresponding derivative of $k h_\xi \cos\theta$. The coupling vanishes in the identity exterior when the medium is also range-independent there.

Higher-order directional diagonalization. A higher-order one-way symbol can be constructed by eliminating the off-diagonal coupling order by order. To state the recursion, write $A_+ = i\Lambda + T$ and $A_- = -i\Lambda + T$ and represent a locally forward subspace as a graph $u_- = K(\xi) u_+$. Invariance of this graph under (15) requires the operator Riccati equation

$$K_\xi = \mathcal{R} + A_- K - K A_+ - K \mathcal{R} K. \tag{23}$$

If k is the Weyl symbol of K, this becomes the symbol Riccati equation

$$k_\xi = \rho + a_-\#k - k\#a_+ - k\#\rho\#k, \qquad a_\pm = \sigma(A_\pm). \tag{24}$$

The leading symbol equation has denominator $2i\lambda$; it is solvable as long as the forward and backward sheets remain separated. Expanding K in powers of ε and substituting back into the u_+ equation produces successive local corrections to a one-way generator. This is the pseudodifferential counterpart of higher-order WKB factorization [4, 5]. It does not replace the nonlocal two-way solution near a turning point or a strong reflector.

6 Bremmer series and boundary-value formulation

Let $G_+(\xi, s)$ and $G_-(\xi, s)$ be evolution operators for the diagonal generators A_+ and A_-:

$$\partial_\xi G_\pm(\xi, s) = A_\pm(\xi) G_\pm(\xi, s), \qquad G_\pm(s, s) = I. \tag{25}$$

For real self-adjoint B, these propagators are unitary on the propagating subspace. Given an incident forward field at $\xi = 0$ and no incoming backward field at $\xi = X$, the exact variation-of-constants equations are

$$u_+(\xi) = G_+(\xi, 0)u_+(0) + \int_0^\xi G_+(\xi, s)\mathcal{R}(s)u_-(s)\, ds, \tag{26}$$

$$u_-(\xi) = -\int_\xi^X G_-(\xi, s)\mathcal{R}(s)u_+(s)\, ds. \tag{27}$$

These equations form a two-point boundary-value problem. Substitution of (27) into (26), followed by iteration, yields the Bremmer series. Each additional round trip introduces two interactions with $\mathcal{R}$ and one additional pair of directional propagations. A first backward approximation follows by substituting the zeroth forward field $u_+^{(0)}(s) = G_+(s, 0)u_+(0)$ into (27):

$$u_-^{(1)}(\xi) = -\int_\xi^X G_-(\xi, s)\mathcal{R}(s)G_+(s, 0)u_+(0)\, ds. \tag{28}$$

For a single propagating mode with phase $\Phi_\xi = \lambda > 0$, the first reflection has the oscillatory form

$$u_-^{(1)}(\xi) = -e^{-i\Phi(\xi)} \int_\xi^X \rho(s)e^{2i\Phi(s)}u_+(0)\, ds. \tag{29}$$

Integration by parts separates a locally slaved component, of relative size $O(\varepsilon)$, from a rapidly oscillating reflected remainder. Feeding the local component back into the forward equation produces order-ε^2 corrections, including the familiar second-order WKB terms. Genuine backscatter is controlled by the regularity of the range variation and the separation of the two characteristic sheets. With C^m coefficients, compactly supported range variation, and no stationary phase point, repeated integration by parts gives algebraic decay with the available number of derivatives. A C^∞ profile gives superalgebraic decay; an exponential estimate requires analytic continuation and corresponding contour-deformation hypotheses. A merely C^2 terrain map does not justify an $O(\varepsilon^\infty)$ or exponential claim.

For the operator series, a sufficient small-coupling condition is

$$\int_0^X \|\mathcal{R}(s)\|\, ds < 1. \tag{30}$$

Under unitary diagonal propagation, this bounds successive integral iterations. It is sufficient rather than necessary; near grazing, a turning point, a sharp interface, or a strong reflector the series may require a different preconditioner or a direct block boundary solve.

7 Weyl-symbol propagators and rational approximation

7.1 Symbol evolution and Bremmer terms

Let $\mathbf{u} = (u_+, u_-)^T$ and let $\mathbf{a}$ be the matrix Weyl symbol of the generator in (15):

$$\mathbf{a} = \begin{pmatrix} a_+ & \rho \\ \rho & a_- \end{pmatrix}, \qquad a_+ = i\lambda + t, \quad a_- = -i\lambda + t, \tag{31}$$

where $\lambda = \sigma(\Lambda)$, $t = \sigma(T)$, and $\rho = \sigma(\mathcal{R})$. For a 2×2 symbol matrix, $\#$ denotes matrix multiplication with Weyl products in the entries. The propagator symbol $\mathbf{g}(\xi, s)$ is defined by

$$\partial_\xi \mathbf{g}(\xi, s) = \mathbf{a}(\xi)\#\mathbf{g}(\xi, s), \qquad \mathbf{g}(s, s) = \mathbf{I}. \tag{32}$$

Thus the exact symbol solution is a range-ordered *star exponential*, not the pointwise exponential of the integrated symbol. At leading semiclassical order, $\#$ becomes ordinary multiplication and

$$\partial_\xi \mathbf{g}_0 = \mathbf{a}_0 \mathbf{g}_0. \tag{33}$$

The next terms follow by substituting the Moyal expansion (7) into (32); they include phase-space derivatives and account for noncommutation of the vertical operators at different ranges.

The diagonal propagator symbols $g_\pm(\xi, s)$ satisfy

$$\partial_\xi g_\pm(\xi, s) = a_\pm(\xi)\# g_\pm(\xi, s), \qquad g_\pm(s, s) = 1. \tag{34}$$

The first reflected field is the operator with symbol

$$s_-^{(1)}(\xi) = -\int_\xi^X g_-(\xi, s)\#\rho(s)\#g_+(s, 0)\, ds, \qquad u_-^{(1)}(\xi) = \mathrm{Op}^w(s_-^{(1)}(\xi))u_+(0). \tag{35}$$

Higher Bremmer orders are nested range integrals of alternating g_+ and g_- symbols, with a factor ρ at every change of direction and with all products interpreted as $\#$. For a single isolated ray, the leading star products reduce to scalar products and (35) reduces to the oscillatory reflection integral (29).

7.2 Frozen-range propagator

For a short step $d = \Delta\xi$, first freeze the coefficients at the midpoint $\xi_{j+1/2}$. The diagonal operator generators are $A_\pm = \pm i\Lambda + T$. A symmetric splitting separates the skew-adjoint commutator correction from the principal phase:

$$G_\pm(\xi_{j+1}, \xi_j) \simeq e^{dT_{j+1/2}/2}\, e^{\pm id\Lambda_{j+1/2}} e^{dT_{j+1/2}/2} + O(d^3), \tag{36}$$

for frozen smooth coefficients. A higher-order Magnus or symmetric composition can be used when the range variation over one step is not negligible. The middle factor is a function of the vertical operator. Choose a reference wavenumber $\beta > 0$ and define

$$L_\beta = \frac{B}{\beta^2} - I, \qquad \ell_\beta(\xi, \eta, q) = \frac{b(\xi, \eta, q)}{\beta^2} - 1. \tag{37}$$

Then

$$e^{\pm id\Lambda} = e^{\pm i\beta d} \exp\left[\pm i\beta d \left(\sqrt{I + L_\beta} - I\right)\right]. \tag{38}$$

This is the same reduced square-root phase used in Lytaev's JCS22 one-way propagator [6], with L_β now the Liouville-transformed terrain-following vertical operator.

7.3 Rational approximation in Weyl form

Let $\mathscr{R}_\pm(\zeta)$ be a rational approximation on the propagating spectral interval $\mathcal{I}$ to

$$F_\pm(\zeta) = \exp\Big[\pm i\beta d\left(\sqrt{1+\zeta}-1\right)\Big], \qquad \zeta \in \mathcal{I}. \tag{39}$$

For example, in factored form,

$$\mathscr{R}_\pm(\zeta) = \prod_{m=1}^{p} \frac{1+a_m^\pm \zeta}{1+b_m^\pm \zeta}, \qquad \sup_{\zeta\in\mathcal{I}} |F_\pm(\zeta) - \mathscr{R}_\pm(\zeta)| \le \delta. \tag{40}$$

The coefficients may come from a local Padé fit about $\zeta = 0$ or from an interval fit such as Chebyshev–Padé or rational interpolation. The interval is selected from the desired angle and wavenumber aperture, as in JCS22. In terrain-following coordinates its endpoints must include the metric-scaled spectrum $b_0 \simeq k^2 h_\xi^2 - q^2/r^2$ (and the intended subprincipal corrections if they are retained).

At the operator level, set

$$\mathscr{R}_\pm(L_\beta) = \prod_{m=1}^{p} (I + a_m^\pm L_\beta) \left[\prod_{m=1}^{p} (I + b_m^\pm L_\beta)\right]^{-1}. \tag{41}$$

Each factor requires one shifted solve. For example, $(I + b_m^\pm L_\beta)v_m = (I + a_m^\pm L_\beta)w_{m-1}$. With the flux-form finite-difference realization of B, these are the tridiagonal vertical solves already used by the high-order terrain-following propagator. The two directions use conjugate rational fits for real B, so the discrete frozen-range phase pair has the expected adjoint relation.

The exact Weyl symbol of a rational function is most cleanly written using star products. For a polynomial $p(\zeta) = \sum_j p_j \zeta^j$, define $p^\#(\ell_\beta) = \sum_j p_j \ell_\beta^{\#j}$, with $\ell_\beta^{\#0} = 1$ and $\ell_\beta^{\#(j+1)} = \ell_\beta^{\#j} \# \ell_\beta$. If $p_\pm(\zeta)$ and $q_\pm(\zeta)$ are the numerator and denominator polynomials in (40), form their star-functional-calculus symbols $p_\pm^\#(\ell_\beta)$ and $q_\pm^\#(\ell_\beta)$. Then

$$\sigma^w(\mathscr{R}_\pm(L_\beta)) = p_\pm^\#(\ell_\beta) \# \left(q_\pm^\#(\ell_\beta)\right)_\#^{-1}, \qquad q_\pm^\#(\ell_\beta) \# \left(q_\pm^\#(\ell_\beta)\right)_\#^{-1} = 1. \tag{42}$$

The star inverse exists on the chosen symbol region when the rational denominator has no zero there. Since the first Weyl correction to a scalar function of one real symbol vanishes, the principal symbol is simply

$$\sigma^w(\mathscr{R}_\pm(L_\beta)) = \mathscr{R}_\pm(\ell_\beta) + O(\varepsilon^2). \tag{43}$$

For the commutator half-step write $\tau_d = \sigma^w(e^{dT/2})$; at principal order $\tau_d = e^{dt/2} + O(\varepsilon^2)$. Combining (36)–(43) gives the frozen-range Weyl symbol of the rationalized directional propagator:

$$g_{\pm,j}^{\mathrm{rat}} = \tau_d \# \left[e^{\pm i\beta d} p_\pm^\#(\ell_\beta) \# \left(q_\pm^\#(\ell_\beta)\right)_\#^{-1}\right] \# \tau_d + O(d^3), \tag{44}$$

where all symbols are evaluated at $\xi_{j+1/2}$. At principal order,

$$g_{\pm,j}^{\mathrm{rat}} \simeq \exp\Big[\pm id\sqrt{b(\xi_{j+1/2},\eta,q)}\Big], \qquad |\text{phase error}| \le \delta + O(\varepsilon^2) + O(d^3). \tag{45}$$

The local Padé fit concentrates accuracy near normal incidence ($\ell_\beta = 0$); an interval fit controls the error over the full symbol support. This symbol-level error is distinct from the vertical finite difference dispersion error and from the Bremmer-series truncation error.

In flat Cartesian coordinates $h_\xi = h_\eta = r = 1$, let the local wavenumber be $k_0 n(\xi, \eta)$ and choose reference $\beta = k_0$. Then $b = k_0^2 n^2 - q^2$ and

$$\ell_{k_0} = -\frac{q^2}{k_0^2} + (n^2 - 1). \tag{46}$$

For $n = 1$ this reduces to $-q^2/k_0^2$, exactly the homogeneous-medium case; for variable n it is the scalar argument used in JCS22. Equation (41) is therefore its terrain-following operator generalization, while (42)–(45) give the corresponding Weyl-symbol derivation.

8 Discrete realization in terrain-following coordinates

8.1 Vertical operator and one-way building blocks

On a range grid $\xi_j = j\Delta\xi$ and vertical grid $\eta_n = n\Delta\eta$, construct a matrix B_j from the flux-form operator (5). The discrete vertical scheme should preserve the self-adjoint weighting and boundary conditions. In the existing terrain-following JAX propagator, the coefficient arrays include the metric weight $m = r^2$, the flux coefficient r (or its density-weighted version), the local wavenumber contrast, and midpoint flux coefficients. Each rational substep is a tridiagonal solve; the range step also includes a flux-normalizing fourth-root pair [8].

Those pieces can be reused to build actions of B_j, and rational resolvent solves can approximate $B_j^{1/2}$ and $B_j^{1/4}$ on a specified propagating spectrum. Alternatively, for small verification problems, a symmetric matrix eigendecomposition gives a reference implementation. Define

$$Q_j = B_j^{1/4}, \qquad \Lambda_j = B_j^{1/2}, \qquad \mathcal{R}_{j+1/2} = \frac{1}{2}\left(\frac{Q_{j+1} - Q_j}{\Delta\xi} Q_{j+1/2}^{-1} + Q_{j+1/2}^{-1} \frac{Q_{j+1} - Q_j}{\Delta\xi}\right), \tag{47}$$

with a consistent midpoint or higher-order approximation for $Q_{j+1/2}$. The finite difference in (47) approximates Q_ξ; for high accuracy it should be replaced by a symmetric multi-column differentiation or a direct discrete derivative of the matrix-function action. A discontinuous staircase metric is unsuitable for this derivative because it creates artificial impulses in the coupling.

8.2 Two directional sweeps

Let $\mathcal{P}_+$ and $\mathcal{P}_-$ be discrete one-way propagators for A_+ and A_-, including the skew-adjoint term T to the desired order. Given a current forward field, a backward sweep evaluates the discrete version of (27); the next forward sweep evaluates (26). Starting with $u_-^{(0)} = 0$ gives the successive Bremmer reflection corrections. An equivalent approach solves the block tridiagonal discretization of (15) directly, which avoids fixed-point convergence issues and provides a reference for truncated-series results.

Component	Existing terrain-following code	Required two-way extension
Map and metric	Orthogonal map, h_ξ, h_η, r	Reuse unchanged
Vertical operator	High-order flux-form rational solves	Expose symmetric B_j action
Range step	Forward rational PE step	Separate G_+ and G_-
Flux scaling	Approximate $B^{\pm 1/4}$ column pair	Root actions in the split state
Backscatter	Not represented	$\mathcal{R}$ and backward range sweep
Range boundary	One-way source/outgoing top	Incident data at 0, no incoming at X

8.3 Boundary conditions and absorbing regions

The terrain boundary condition is imposed on the coordinate line $\eta = 0$. For a finite-impedance boundary, the physical normal derivative must be transformed with the local metric; the current solver's compact closure is a candidate discretization for this part. At the top, the orthogonal map is the identity, so a vertical outgoing condition can be imposed without map corrections. For the range boundary, one prescribes the incident $u_+(0)$ and sets the incoming-from-the-right component $u_-(X)$ to zero. This range condition is distinct from the vertical top boundary condition.

An absorbing layer makes the discrete operator non-self-adjoint. Then the flux conservation and unitary bounds above no longer hold exactly. A robust implementation should first validate the two-way solver on a lossless finite domain with exact outgoing vertical conditions. Absorption can then be added with a monitored reflection coefficient and a biorthogonal or energy estimate appropriate to the damped operator.

9 Verification programme

A two-way implementation should be developed in stages.

1. **Constant flat medium.** Check that $\mathcal{R} = 0$, the two components propagate with the expected signs, and their superposition agrees with the exact Helmholtz solution.

2. **Range-dependent one-dimensional medium.** With no vertical variation, compare the full two-way result and successive Bremmer orders against an exact transfer-matrix or boundary-value solution. Measure both the reflected amplitude and convergence with range step.

3. **Smooth terrain map.** Use a smooth isolated hill and compare zero-, first-, and higher-reflection results with a direct two-way discretization. Refine the map and grid independently to separate terrain regularity error from propagation error.

4. **Impedance and evanescence.** Add finite ground impedance, then evanescent modes and absorption, checking flux balance where applicable and boundary reflection where flux conservation does not apply.

No numerical accuracy or speed claim is made here; the existing one-way JAX solver is a source of reusable discretization components, not a two-way reference solution.

10 Conclusion

Orthogonal terrain-following coordinates put the terrain-boundary Helmholtz equation in a form suited to directional operator factorization. The Liouville-transformed equation has a symmetric vertical operator whose fractional powers define flux-normalized forward and backward components. Their exact coupled evolution yields a Bremmer series, while Weyl calculus identifies the leading interaction and its higher-order corrections.

The existing high-order terrain-following PE contributes the coordinate map, metric and flux-form vertical solves. A genuine two-way higher-order propagator still requires discrete square-root actions, consistent coupling operators, and a two-point range boundary solution. Deriving and validating those components is the next step; reversing the existing one-way march alone does not produce a Bremmer solver.

References

[1] H. Bremmer, "The W.K.B. approximation as the first term of a geometric-optical series," *Communications on Pure and Applied Mathematics*, vol. 4, pp. 105–115, 1951.

[2] M. V. de Hoop, "Generalization of the Bremmer coupling series," *Journal of Mathematical Physics*, vol. 37, no. 7, pp. 3246–3282, 1996, doi: 10.1063/1.531566.

[3] A. K. Gautesen and M. V. de Hoop, "Uniform asymptotic expansion of the generalized Bremmer series," *SIAM Journal on Applied Mathematics*, vol. 60, no. 4, pp. 1302–1329, 2000, doi: 10.1137/S0036139997323619.

[4] L. Fishman and J. J. McCoy, "Derivation and application of extended parabolic wave theories. I. The factorized Helmholtz equation," *Journal of Mathematical Physics*, vol. 25, no. 2, pp. 285–296, 1984.

[5] C. C. Stolk, "A pseudodifferential equation with damping for one-way wave propagation in inhomogeneous acoustic media," *Wave Motion*, vol. 40, no. 2, pp. 111–121, 2004, doi: 10.1016/j.wavemoti.2003.12.016.

[6] M. S. Lytaev, "Rational interpolation of the one-way Helmholtz propagator," *Journal of Computational Science*, vol. 58, article 101536, 2022, doi: 10.1016/j.jocs.2021.101536.

[7] M. Zworski, *Semiclassical Analysis*. Providence, RI: American Mathematical Society, 2012.

[8] M. S. Lytaev, "Orthogonal terrain-following coordinates for the higher-order parabolic equation: Wide-angle propagation over irregular terrain with discrete transparent boundaries," manuscript accompanying the terrain-following implementation, `pywaveprop/helmholtz_tf_jax.py`.